YIPING JI

Google Scholar ◇ LinkedIn ◇ Personal Page ◇ Email

RESEARCH INTERESTS

My research focuses on the mathematical foundations, optimization, and scalable training of foundation models, particularly transformer-based architectures. I am also interested in representation learning in various modeling.

EDUCATION

University of Adelaide *Oct 2023 - present*
Ph.D. in Computer Science and Mathematics (supervisor: Prof. Simon Lucey) *Adelaide, Australia*
- Research into unveiling the mathematical foundations underlying the processes of transformer models.

Tel Aviv University *Oct 2019 - July 2023*
M.Sc. in Applied Mathematics (supervisor: Prof. Nir Sochen) *Tel Aviv, Israel*
- Master Thesis: unsupervised learning optical flow using implicit neural representation

Tel Aviv University *Oct 2015 - July 2019*
B.Sc in Electrical and Electronic Engineering *Tel Aviv, Israel*

RESEARCH EXPERIENCE

KV Cache Compression *Oct 2026 – Present*

- Developed KV merging algorithms to prevent performance degradation in large compression rate.

Representation Learning *Apr 2024 – Present*

- Investigated how architecture design, particularly residual connections, and optimizer choice shape the representation in discriminative vision models, large language models and generative models.

Transformer-Based Architecture and Training *Apr 2024 – Present*

- Developed conditioning-aware initialization and optimization methods for training fully residual-free Transformers, enabling stable convergence without conventional skip connections for the first time.
- Investigated how self-attention, normalization layers and residual connections affect optimization, scalability.

Efficient Adaptation and Model Compression *Oct 2023 – Present*

- Developed sine-activated low-rank matrices that increase effective rank while preserving parameter efficiency, improving adaptation across language models, Vision Transformers, and Diffusion Models. The algorithm is internally in use by industry partner.

- Studied post-training quantization and model compression, including the quantization advantages of residual-free Transformer architectures.

SELECTED PUBLICATION LIST

- **Should We Skip Diffusion?**
  Y. Ji, J. Martens, S. Lucey
  In Arxiv.[Paper]

- **The Quantization Benefits of Residual-Free Transformers**
  Y. Ji*, M. Sabanayagam*, P. Moghadam, H. Saratchandran, S. Lucey. (*Equal contribution)
  In *Neural Information Processing Systems* ***(NIPS)***, 2026.[Paper]
- **Cutting the Skip: Training Residual-Free Transformers**
  Y. Ji, J. Martens, J. Zheng, Z. Zhou, P. Moghadam, X. Zhang, H. Saratchandran, S. Lucey.
  In *International Conference on Learning Representation* ***(ICLR)***, 2026. [Paper]
- **Compressing Sine-Activated Low-Rank Adapters through Post-Training Quantization**
  C. Gordon*, **Y. Ji***, H. Saratchandran*, P. Albert, S. Lucey. (*Equal contribution)
  In *Proceedings of the AAAI Conference on Artificial Intelligence* ***(AAAI)****, 2026* [Paper]
- **Always Skip Attention**
  Y. Ji, H. Saratchandran, P. Moghadam, S. Lucey.
  In *International Conference on Computer Vision* ***(ICCV)***, 2025. [Paper]
- **Rethinking Softmax: Self-Attention with Polynomial Activations**
  H. Saratchandran, J. Zheng, **Y. Ji**, W. Zhang, S. Lucey.
  In *Neural Information Processing Systems* ***(NIPS)***, 2026.[Paper]
- **Efficient Learning with Sine-Activated Low-Rank Matrices**
  Y. Ji*, H. Saratchandran*, C. Gordon, Z. Zhang, S. Lucey. (*Equal contribution)
  In *International Conference on Learning Representation* ***(ICLR)***, 2025. [Paper]
  (Internally used by industrial partner)

AWARDS AND SCHOLARSHIPS

Google Cloud Research Credits Program *Dec 2025*
DATA61 PhD top-up Scholarships , CSIRO, Australia *Mar 2024 - present*
Full Fee Scholarships , University of Adelaide, Australia *Oct 2023 - present*
Full Government Scholarships , Tel Aviv University, Israel *2015 - 2019*

SERVICE

Reviewer: ICLR 2025,2026, AAAI 2026, ECCV 2026, NIPS 2026

REFERENCES

Available upon request.